

CUSTOMER CHURN ANALYSIS

Telecom Industry | Python Exploratory Data Analysis

Key Insight

26.54% of customers have churned — this report uncovers why.

1. Executive Summary

This report presents an Exploratory Data Analysis (EDA) of a telecom company's customer churn dataset using Python. The analysis identifies key drivers of churn and provides actionable insights to help the business improve customer retention. The dataset contains demographic information, account details, and service usage records for thousands of customers.

Metric	Value	Interpretation
Overall Churn Rate	26.54%	About 1 in 4 customers has churned
Highest Risk Segment	Month-to-Month Contracts	Highest churn among all contract types
Senior Citizen Churn	Comparatively Higher	Seniors churn more than non-seniors
Riskiest Payment Method	Electronic Check	Most churned customers use e-check
Low Tenure Churn	1–2 Months	New customers are most likely to leave

2. Dataset Overview

The dataset used for this analysis is the Telecom Customer Churn dataset (Customer Churn.csv). Below is a summary of the key data characteristics and preprocessing steps performed.

2.1 Data Characteristics

Attribute	Details
File	Customer Churn.csv
Target Variable	Churn (Yes / No)
Key Features	Demographics, Services, Contract, Payment
Data Types	Categorical and Numerical
Missing Values	None after cleaning
Duplicates	None found

2.2 Preprocessing Steps

- TotalCharges column: Blank values replaced with '0' and converted from object to float dtype.
- TotalChanges column: Dropped as it was redundant / erroneous.
- SeniorCitizen column: Integer values (0/1) converted to human-readable labels ('Yes'/'No') for better interpretability in charts.
- No duplicate records were found across the dataset.
- No null values remained after cleaning.

3. Exploratory Data Analysis (EDA)

3.1 Overall Churn Distribution

A count plot and pie chart were generated to visualize the overall churn distribution. The analysis revealed:

- Total customers who did NOT churn: 73.46%
- Total customers who DID churn: 26.54%

This confirms that roughly one in four customers left the company — a significant retention problem that demands attention.

3.2 Churn by Gender

A countplot segmented by gender and churn was created. The key finding: churn rate is approximately equal between male and female customers. Gender does not appear to be a significant predictor of churn in this dataset.

3.3 Churn by Senior Citizen Status

Analysis of senior citizen status revealed a noteworthy pattern:

- Senior citizens (SeniorCitizen = Yes) have a comparatively higher churn rate than non-seniors.
- Non-senior customers make up the majority of the customer base.
- A stacked bar chart with percentage labels confirmed this disparity clearly.

This suggests senior customers may require specialized retention strategies, such as dedicated support or simplified plans.

3.4 Churn by Tenure

A histogram of tenure (months with the company) segmented by churn was plotted. The insight:

- Customers who have been with the company for only 1–2 months are the most likely to churn.
- Long-tenure customers (e.g., 60+ months) show strong loyalty and very low churn.

This suggests the critical retention window is within the first few months of a customer's lifecycle. Early engagement programs and onboarding improvements could significantly reduce churn.

3.5 Churn by Contract Type

A countplot of contract type versus churn was generated. The results show a clear pattern:

Contract Type	Churn Likelihood	Observation
Month-to-Month	High	Customers have no long-term commitment; easiest to leave
One Year	Medium	Moderate commitment reduces churn significantly
Two Year	Low	Strong commitment; very few churn

Recommendation: Incentivize customers to switch from month-to-month to annual contracts through discounts or loyalty rewards.

3.6 Churn by Services Used

A 3x3 grid of countplots was created for 9 service-related features: PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, and StreamingMovies.

Key findings:

- Customers without OnlineSecurity, TechSupport, and OnlineBackup have noticeably higher churn rates.
- DSL internet service customers churn less than fiber optic customers.
- Customers with PhoneService are generally retained better.
- StreamingTV and StreamingMovies did not show significant differences in churn rates.

Recommendation: Bundle value-added services like OnlineSecurity and TechSupport proactively — customers who use more services stay longer.

3.7 Churn by Payment Method

A countplot of payment method segmented by churn was analyzed. The results:

Payment Method	Churn Pattern	Implication
Electronic Check	Highest Churn	Most at-risk payment segment
Mailed Check	Moderate Churn	Somewhat at risk
Bank Transfer (Auto)	Lower Churn	Auto-pay customers are more stable
Credit Card (Auto)	Lower Churn	Auto-pay customers are more stable

Recommendation: Encourage customers to switch to automated payment methods (bank transfer or credit card auto-pay), which correlate strongly with retention.

4. Key Findings Summary

#	Finding	Business Impact
1	26.54% overall churn rate	High — nearly 1 in 4 customers lost
2	New customers (1-2 months) churn most	Critical onboarding period needs improvement
3	Month-to-month contracts drive churn	Push for longer contract commitments

4	Senior citizens churn more	Need tailored retention programs for seniors
5	Electronic check users churn most	Promote auto-pay adoption
6	No OnlineSecurity/TechSupport = higher churn	Bundle protective services proactively
7	Gender does not affect churn	No need for gender-based retention strategy
8	Long-tenure customers are loyal	Reward loyalty to maintain retention

5. Recommendations

5.1 Immediate Actions

- **Launch an early engagement program targeting customers in their first 1–3 months to reduce new customer churn.**
- **Offer discounts or incentives to convert month-to-month customers to annual contracts.**
- **Promote automatic payment methods (bank transfer / credit card) to reduce churn risk.**

5.2 Medium-Term Strategies

- Bundle OnlineSecurity and TechSupport services into default packages or offer them as low-cost add-ons.
- Create a senior citizen loyalty program with simplified interfaces, dedicated support, and special pricing.
- Implement a customer health score model to proactively identify at-risk customers before they churn.

5.3 Long-Term Initiatives

- Build a machine learning churn prediction model using features identified in this EDA.
- Establish quarterly churn reviews to track trends and measure the impact of retention strategies.
- Invest in customer satisfaction surveys at key lifecycle milestones (1 month, 6 months, 1 year).

6. Tools & Methodology

Tool / Library	Version	Purpose
Python	3.x	Core programming language
Pandas	Latest	Data loading, cleaning, and manipulation
NumPy	Latest	Numerical operations
Matplotlib	Latest	Base plotting and chart rendering
Seaborn	Latest	Statistical visualizations (countplots, histplots)
Jupyter Notebook	Latest	Interactive analysis environment

Methodology: This analysis follows a standard EDA workflow — data ingestion, cleaning and validation, univariate analysis, bivariate analysis (features vs. churn), and insight generation. All visualizations were created inline within the Jupyter Notebook environment.

7. Conclusion

This Customer Churn Analysis successfully identified the primary drivers of customer attrition in the telecom dataset. With a churn rate of 26.54%, the company faces a significant retention challenge. However, the analysis reveals clear, actionable patterns:

- Short-tenure, month-to-month, electronic check customers are most at risk.
- Value-added services (security, support) are strongly associated with retention.
- Automated payments and longer contracts correlate with customer loyalty.

By targeting these specific segments with tailored retention strategies, the company can make meaningful progress in reducing churn and improving long-term customer lifetime value.

— End of Report —

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